

Unit-3 Buckling of Column's:-

①

Compressible member's of relatively long length which fail by buckling are called column's.

Column's are compression member's that are slender when compared to its length, and under gradually increasing load fail by buckling.

When a member is subjected to compressive forces, depending upon the ratio of length to lateral dimension it may fail due to,

(a) Crushing - without buckling - short column's.

(b) A combination of crushing and buckling - Intermediate column's.

(c) failure due to buckling (lateral bending) - long column's.

Hence, there is a definite load called 'Critical load' or otherwise also called 'Crushing' load ' P_c ', at which a long column will fail by buckling.

The failure occurs by a sudden excessive lateral deflection, The lateral deflection remains as long as the load is applied and when the load is removed, the specimen returns back to its original shape. Hence, buckling phenomenon is an elastic instability.

Slenderness Ratio

(2)

The ratio of length to the radius of gyration is called slenderness ratio.

Where radius of gyration is defined / described as

$$k = \sqrt{\frac{I}{A}} \quad \text{where } I \text{ is the Area moment of Inertia}$$

and A is the Cross sectional Area.

Euler's theory of Buckling failure:-

Consider a slender column with pinned ends, The column is loaded vertically through the Centroid of the Cross-section without any eccentricity.

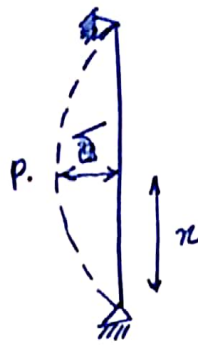
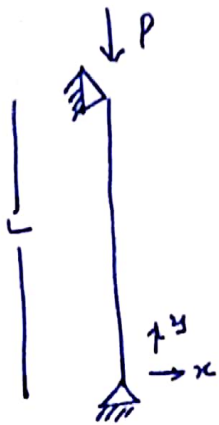
Assumptions Made:-

- ① Column is perfectly straight.
- ② Material is homogenous, linear, elastic and follows Hooke's law.
- ③ The strength to resist buckling is affected by end conditions.
- ④ Flexural rigidity is Uniform.

When the axial load P is small, the column remains straight and undergoes only axial compression, which is Uniform. Thus we can say the column is in a stable equilibrium condition.

As, the load P is gradually increased, we reach a condition of neutral equilibrium, in which the column has a bent shape.

(3)



The lateral deflection δ is so small that there is no appreciable difference between Original length of the Column.

Given Below is the Euler's Curve for failure.

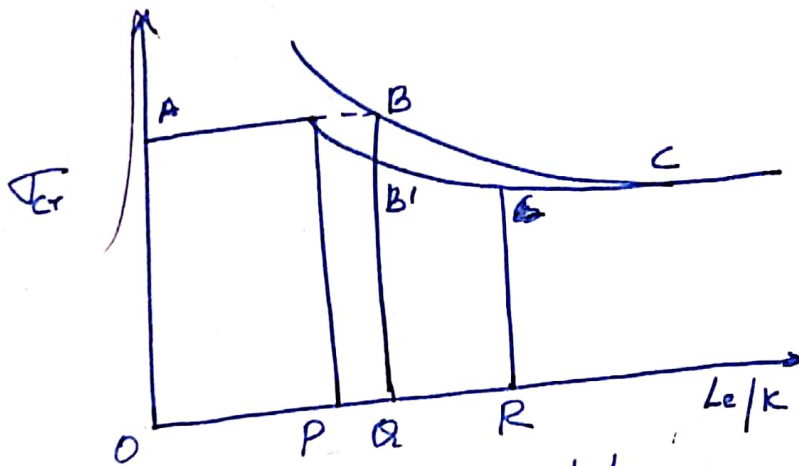


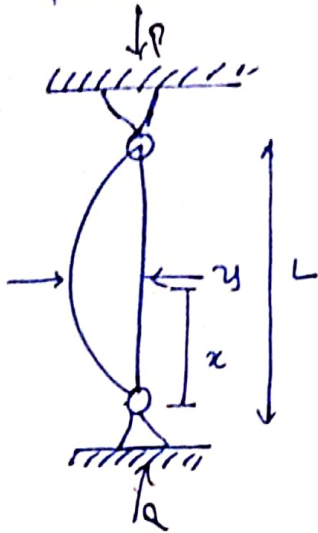
fig shows graph of Critical load vs radius of gyration.

is called Euler's hyperbola,
 $AB'C$ is the graph obtained from experimental data.
 OP is Short Columns with low Slenderness ratio.
 PR is Intermediate members with medium Slenderness ratio.
 QR is long columns/members with high Slenderness ratio.

In the next sections, we will derive the Euler's critical/crippling load for a column with vary End conditions.

- (a) Both ends hinged, (b) Both sides fixed
- (c) One side fixed other End free, (d) One End fixed other ends hinged.

Both Ends Hinged:-



The bending Moment at Section at a distance of x from the base

$$EI \cdot \frac{d^2 y}{dx^2} = -Py.$$

$$EI \frac{d^2 y}{dx^2} + Py = 0, \quad \frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0.$$

The solution to the differential equation is given by.

$$y = C_1 \cos(x\sqrt{P/EI}) + C_2 (\sin(x\sqrt{P/EI})).$$

Where C_1 and C_2 are Constants of Integration.

Applying Boundary Conditions and Solving.

$x=0, y=0$, we get $C_1=0$.

$x=L, y=0$,

$$0 = 0 + C_2 \sin(L\sqrt{P/EI})$$

Hence, $C_2 \sin(L\sqrt{P/EI}) = 0$. We cannot take $C_2 = 0$ as the entire equation will become zero.

$$\therefore \sin(L\sqrt{P/EI}) = 0.$$

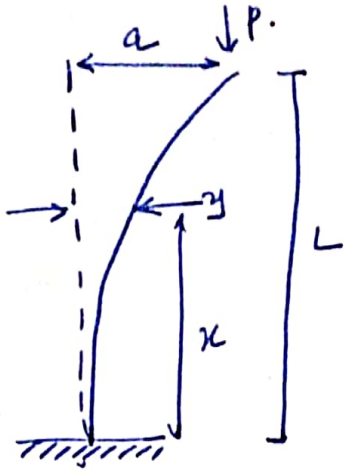
$L\sqrt{\frac{P}{EI}} = 0$, rearranging the terms, we get,

$$P = \frac{\pi^2 EI}{L^2}$$

This load is called Critical or buckling load, sometimes also called Crippling load.

One End fixed and other End free:

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Similarly obtaining the bending equation.

$$EI \frac{d^2y}{dx^2} = P(a-y).$$

Rearranging we get,

$$\frac{d^2y}{dx^2} + \frac{P}{EI} y = \frac{Pa}{EI}.$$

Solution of the differential equation is,

$$y = C_1 \cos(\sqrt{P/EI} x) + C_2 \sin(\sqrt{P/EI} x) + a.$$

Applying boundary conditions.

At $x=0$, $dy/dx = 0$, differentiating the solution of d.e.

$$dy/dx = -C_1 \sqrt{P/EI} \sin(\sqrt{P/EI} x) + C_2 \sqrt{P/EI} \cos(\sqrt{P/EI} x)$$

Applying B.C's we get,

$$C_2 \sqrt{P/EI} = 0, \quad C_2 = 0.$$

$$\therefore y = C_1 \cos(\sqrt{P/EI} x) + a. \quad y = C_1 \cos(\sqrt{P/EI} x) + a.$$

Applying B.C's, $x=0$, $y=0$, we get, $C_1 + a = 0$.

$0 = C_1 = -a$, replacing C_1 back,

$$y = a - a \cos(\sqrt{P/EI} x).$$

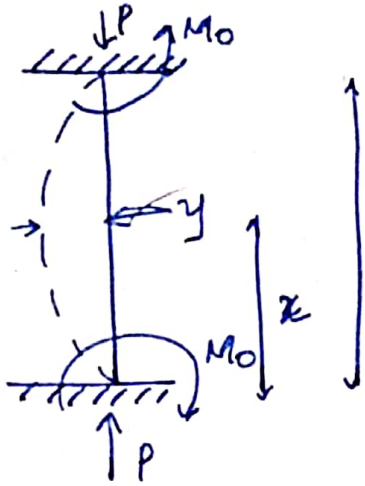
from B.C's, $y=a$ and $x=L$, we get,

$$a \cos(L\sqrt{P/EI}) = 0.$$

Since 'a' cannot be zero,

$$L\sqrt{P/EI} = 0, \quad \text{Rearranging we get, } \boxed{P = \frac{\pi^2 EI}{4L^2}}$$

Both Ends are fixed:-



Deriving the Moment- Equation for the Column,

$$EI \frac{d^2 y}{dx^2} = M_0 - Py$$

$$\frac{d^2 y}{dx^2} + \frac{Py}{EI} = \frac{M_0}{EI}$$

Solution to the differential equation is given by:

$$y = C_1 \cos(\sqrt{P/EI} x) + C_2 \sin(\sqrt{P/EI} x) + M_0/P$$

$$dy/dx = -C_1 \sqrt{P/EI} \sin(\sqrt{P/EI} x) + C_2 \sqrt{P/EI} \cos(\sqrt{P/EI} x)$$

Applying Boundary Conditions,

$$dy/dx = 0, \text{ at } x=0, \text{ we get } C_2 = 0.$$

$$\text{at } x=0, y=0,$$

$$0 = C_1 + M_0/P, \quad C_1 = -M_0/P.$$

$$y = -\frac{M_0}{P} \cos(\sqrt{P/EI} x) + M_0/P.$$

from the Boundary Conditions, $y=0$, at $x=L$, we get,

$$0 = -\frac{M_0}{P} \cos(L\sqrt{P/EI}) + M_0/P$$

$$\cos(L\sqrt{P/EI}) = 1$$

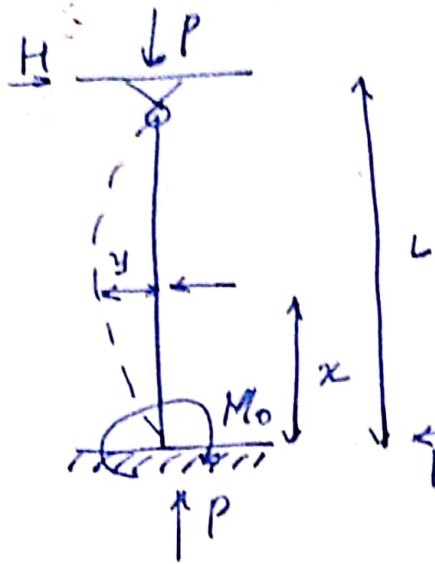
$$L\sqrt{P/EI} = 0, 2\pi, 4\pi, \dots \quad \text{Taking least Significant Value,}$$

$$L\sqrt{P/EI} = 2\pi$$

$$P = 4\pi^2 EI / L^2.$$

One End fixed and other End Hinged:

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The moment at the fixed end is M_0 . The for Equilibrium, the moment has to be $H \times L = M_0$.

Hence, the differential equation is given by,

$$EI \frac{d^2 y}{dx^2} = H(L-x) - Py \quad \left. \begin{array}{l} L \text{ get replaced} \\ \text{by } L-x \text{ for} \\ \text{the section at} \\ \text{distance } x. \end{array} \right\}$$

1. $\frac{d^2 y}{dx^2} = HL - Hx - Py$, rearranging, we get,

$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{H(L-x)}{EI}$, solution for the differential equation is given by,

$$y = C_1 \cos(x\sqrt{P/EI}) + C_2 \sin(x\sqrt{P/EI}) + \frac{H(L-x)}{P}$$

$$dy/dx = -C_1 \sqrt{P/EI} \sin(x\sqrt{P/EI}) + C_2 \sqrt{P/EI} \cos(x\sqrt{P/EI}) - H/P$$

Applying Boundary Conditions,

$x=0, dy/dx=0,$

$$C_2 \sqrt{P/EI} - H/P = 0, \quad C_2 = H/P \sqrt{EI/P}$$

at $x=0, y=0, \quad C_1 + HL/P = 0, \quad C_1 = -HL/P$

at $x=L, y=0, \quad 0 = C_1 \cos(L\sqrt{P/EI}) + C_2 \sin(L\sqrt{P/EI})$

$$\tan L\sqrt{P/EI} = -C_1/C_2 = -[-HL/P] \times P/H\sqrt{P/EI}$$

$$\tan L\sqrt{P/EI} = L\sqrt{P/EI} \quad \left. \begin{array}{l} \text{This resembles } \tan \phi = \phi. \end{array} \right\}$$

$\therefore \phi$ becomes 4.5 radians.

$$L\sqrt{P/EI} = 4.5 \quad \therefore P = \frac{20.25 \cdot EI}{L^2}$$

$$2\pi^2 = 9.85 \times 2 = 19.7 //$$

$$P = \frac{2 \times \pi^2 \cdot EI}{L^2}$$

Effective length:

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We can come up with a general form for all the four cases of Euler's Buckling.

which will be given by.

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} \quad \text{where } L_e \text{ is the effective length.}$$

The effective length is given by,

- ① Both hinged - $L_e = L$
- ② Both fixed - $L_e = L/2$
- ③ One fixed, other free $L_e = 2L$
- ④ One fixed, other hinged $L_e = L/\sqrt{2}$.

Limitations of Euler's buckling Theory:-

The Euler's formulae is based on the assumptions that Column fails only by buckling. This holds good for long slender Column's with high slenderness ratio.

But there are slenderness ratio for which compressive stresses and bending stresses are comparable. In such cases failure will be neither by buckling only nor by crushing only.

Euler's theory is not applicable for such cases, However there are many empirical formulae's based on experimental results. for Example:-

- ① Rankine's formulae
- ② Secant formulae.

Eccentrically Loaded Masonry Column's:-

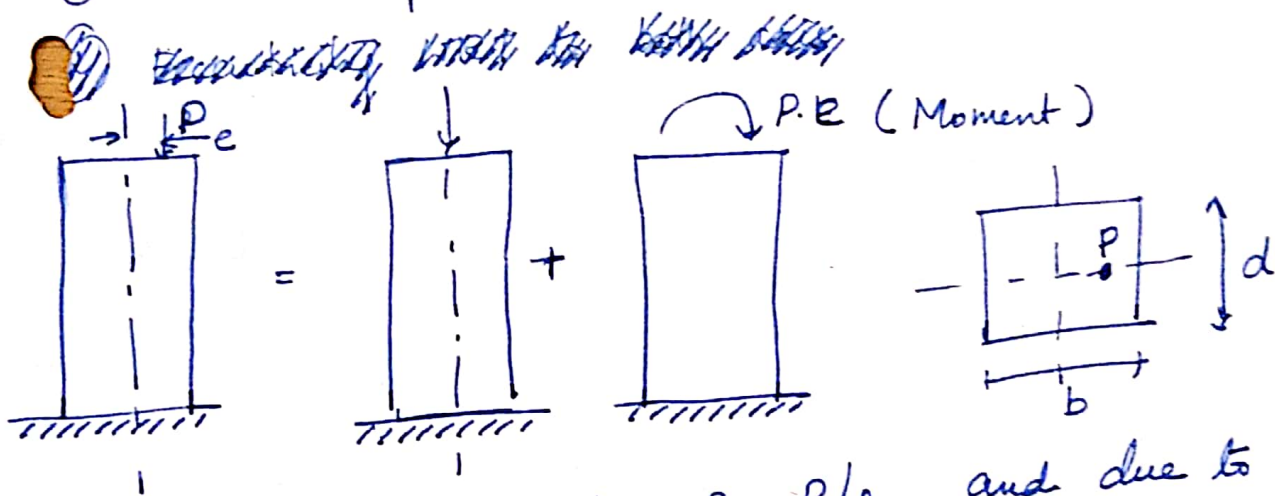
(9)

When a column is loaded eccentrically, it causes bending which produces tension & compression on opposite fibres in the column.

The total stress at any point is the sum of stresses due to axial force and due to bending.

In this we have two cases:-

(i) Eccentricity w.r.t to one axis.



Stresses due to axial force $P_s = P/A$ and due to

moment are $\frac{M \cdot x}{I_y} = \frac{P \cdot e \cdot x}{I_y}$, Hence,

$$\text{max stress} = \frac{P}{A} + \frac{P \cdot e \cdot x}{I_y}$$

$$\text{min stress} = \frac{P}{A} - \frac{P \cdot e \cdot x}{I_y}$$

$$I_y = \frac{1}{12} d b^3 = \frac{1}{12} b^2 A$$

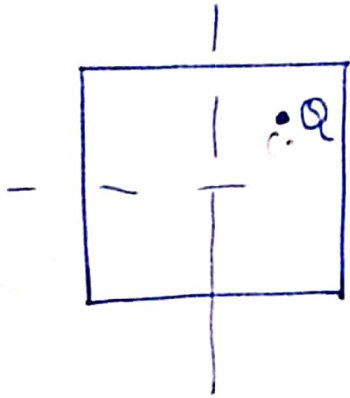
$$\text{Hence, max stress} = \frac{P}{A} + \frac{P \cdot e \cdot b/2}{\frac{1}{12} b^2 A} = \frac{P}{A} \left[1 + \frac{6e}{b} \right]$$

$$\text{and min stress} = \frac{P}{A} \left[1 - \frac{6e}{b} \right]$$

② Eccentricity WRT both axis

10.

Stress at any point- is $\frac{P}{A} + \frac{Pe_1x}{I_y} + \frac{Pe_2y}{I_x}$



When the load/force is applied at point 'Q' on the cross section we have bending in ~~the~~ both the axis.

$$\frac{P}{A} + \frac{Pe_1x}{I_y} + \frac{Pe_2y}{I_x}$$

where e_1 and e_2

are the distance of point Q from the Origin/Centroid.

and x, y are the coordinates of the point at which we need the value of stress.

$$I_y = \frac{1}{12} db^3 = \frac{1}{12} b^2 A$$

$$I_x = \frac{1}{12} d^3 b = \frac{1}{12} d^2 A$$

$$f = \frac{P}{A} \left[1 + \frac{12e_1x}{b^2} + \frac{12e_2y}{d^2} \right]$$

We can substitute the value of x as $b/2$ and y as $d/2$ and find point where the entire load will act as compression.

Numerical Examples:-

(11)

An A-36 steel Column has length of 4m and is pinned at both ends. If the Cross sectional area has the dimensions shown. Determine the Critical load. (Yield stress is 250 MPa) and $E = 200 \text{ GPa}$

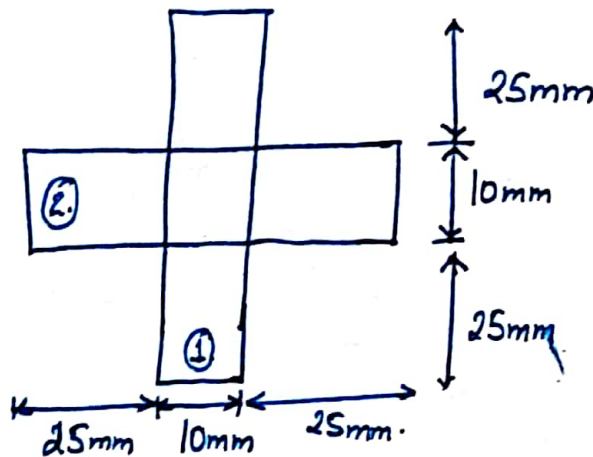
Solution:-

Effective length for both ends pinned is

$$L_e = L$$

Hence, equation reduces to $P_{cr} = \frac{\pi^2 EI}{L^2}$

Cross Section:-



Calculation of I_{xx} & I_{yy} . (Take ~~highest~~ ^{lowest} of the two).

But I_{xx} and I_{yy} are equal due to double symmetry.

$$\begin{aligned} I_{xx} &= I_{xx1} + I_{xx2} = \frac{1}{12} \times 10 \times 60^3 + \frac{1}{12} \times 60 \times 10^3 = \\ &= 180,000 + 5000 = \underline{185000 \text{ mm}^4} \end{aligned}$$

$$P_{cr} = \frac{\pi^2 \times 2.0 \times 10^5 \times 185000}{(4000)^2} = 22.8 \text{ kN.}$$

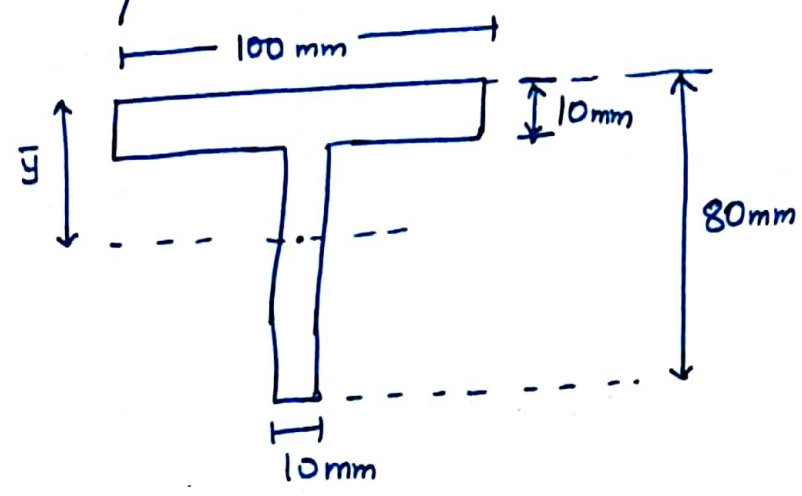
$$\text{Area} = (10 \times 25 \times 4) + (10 \times 10) = 1100 \text{ mm}^2.$$

$$\sigma_{cr} = \frac{22.8 \text{ kN}}{1100 \text{ mm}^2} = 20.7 \text{ MPa} < 250 \text{ MPa. (Hence Euler's formulae is valid)}$$

Example:- 2 :-

Determine the buckling load for a strut of 'Tee-Section' as shown. The strut is 3m long. and hinged at both ends. $E = 200 \text{ GPa}$.

Solution:-



$$\text{Distance of Centroid } \bar{y} = \frac{\sum AY}{\sum A} = \frac{(100 \times 10 \times 5) + (70 \times 10 \times (35 + 10))}{(100 \times 10) + (70 \times 10)}$$

$$\bar{y} = 21.47 \text{ mm.}$$

$$\begin{aligned} I_{xx} &= \sum I_{\bar{x}\bar{x}} + Ay^2 = \frac{1}{12} \times 100 \times 10^3 + (100 \times 10)(21.47 - 5)^2 \\ &\quad + \frac{1}{12} \times 100 \times 70^3 + (10 \times 70)(45 - 21.47)^2 \\ &= 952090 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{yy} &= \sum I_{\bar{y}\bar{y}} = \frac{1}{12} \times 10 \times 100^3 + \frac{1}{12} \times 70 \times 10^3 \\ &= 839166.67 \text{ mm}^4 \end{aligned}$$

Least moment of Inertia is $I_{yy} = 839166.67 \text{ mm}^4$
 Effective length is $l_e = L = 3000 \text{ mm}$.

$$\begin{aligned} P_{cr} &= \frac{\pi^2 EI}{l_e^2} = \frac{\pi^2 \times 200 \times 10^3 \times 839166.67}{(3000)^2} = 184049.85 \text{ N} \\ &= \underline{184.05 \text{ kN}} \end{aligned}$$

Rankine's formulae for buckling and Intermediate column's.

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Euler's formulae is valid only for long column's, i.e. $\frac{L}{K}$ is high

Rankine proposed a formulation for buckling for a very short column to a very large column.

which is,

$$\frac{1}{P_r} = \frac{1}{P_c} + \frac{1}{P_e}$$

Where P_r = Crumbling load

or $P_c = f_c \cdot A = \left\{ \begin{array}{l} \text{Ultimate crushing load for the column} \\ P_c = \sigma_c \cdot A \end{array} \right\}$

and $P_e = \frac{\pi^2 EI}{l^2}$ which is the crumbling load by Euler's formulae.

$$\frac{1}{P_r} = \frac{1}{P_c} + \frac{1}{P_e} = \frac{P_e + P_c}{P_c \cdot P_e} \Rightarrow P_r = \frac{P_c}{1 + \frac{P_c}{P_e}}$$

$$P_r = \frac{\sigma_c \cdot A}{1 + \frac{\sigma_c \cdot A \times l^2}{\pi^2 EI}} = \frac{\sigma_c \cdot A}{1 + \frac{\sigma_c \cdot A \times l^2}{\pi^2 E (AK^2)}} = \frac{\sigma_c \cdot A}{1 + \frac{\sigma_c A l^2}{\pi^2 E AK^2}}$$

$$P_r = \frac{\sigma_c \cdot A}{1 + \frac{\sigma_c}{\pi^2 E} \cdot \left(\frac{l}{K}\right)^2} = \frac{\sigma_c \cdot A}{1 + a \left(\frac{l}{K}\right)^2}, \quad \text{where 'a' is called Rankine's Constant.}$$

Numerical Example:-

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Hollow Cast iron Column 4.5m long with both ends fixed is to carry an axial load of 250kN Under Working Conditions. The internal dia is 0.8 times outer dia of the Column. Using Rankine - Gordon formulae determine diameter of the Column adopting a f.o.s of 4. Assume compressive strength to be 550 N/mm^2 and Rankine's Constant $a = 1/1600$.

Sol:- $l_e = \frac{L}{2} = 2.25 \text{ m} = 2250 \text{ mm}$.

$$d = 0.8D$$

$$\text{Area} = \frac{\pi(D^2 - (0.8D)^2)}{4} = 0.28D^2 \text{ mm}^2$$

$$\text{Moment of Inertia} = I = \frac{\pi(D^4 - (0.8D)^4)}{64} = 0.029 D^4 \text{ mm}^4$$

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{0.029D^4}{0.28D^2}} = 0.32 D \text{ mm}$$

$$\text{Rankine load } P_R = \frac{f_c \cdot A}{1 + a \left(\frac{l_e}{k}\right)^2} = \frac{550 \times 0.28D^2}{1 + \frac{1}{1600} \left(\frac{2250}{0.32D}\right)^2}$$

$$P_R = \frac{154D^2}{1 + 324D} = 250 \times 10^3$$

$$250 \times 10^3 + 81000D = 154D^2$$

$$154D^2 - 250 \times 10^3 - 81000D = 0$$

$D = 1900 \text{ mm}$ and $d = 1520 \text{ mm}$.

Applying factor of safety of 4 we can create a equation.

$$P_R = 154D^2 / (1 + 324D) = 250 \times 10^3 / 4 = 62.5 \times 10^3$$

Numerical Example 2

A round steel rod of diameter 15mm and length 2m is subjected to a gradual increasing axial compressive load. Using Euler's formula to find the buckling load find also the maximum lateral deflection corresponding to the buckling condition. Both ends of the rod may be taken as hinged. $E = 210 \text{ GPa}$ and yield stress of steel $= 240 \text{ N/mm}^2$.

Sol:-

$$A = \frac{\pi d^2}{4} = \frac{\pi \times 15^2}{4} = 176.7 \text{ mm}^2$$
$$I = \frac{\pi d^4}{64} = \frac{\pi 15^4}{64} = 2485 \text{ mm}^4$$

Effective length for both sides hinged is $l_e = 2000 \text{ mm} = L$.

$$P_{cr} = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times 2485}{2000^2} = 1287.61 \text{ N}.$$

Direct Compressive stress, $\sigma_c = P/A = \frac{1287}{176.7} = 7.287 \text{ N/mm}^2$

● $\sigma_c + \sigma_b = \text{Yield stress}.$

$$\sigma_b = \text{Yield stress} - \sigma_c = 240 - 7.287 = 232.71 \text{ N/mm}^2$$

$$\frac{M}{I} = \frac{\sigma_b}{r} = \frac{\sigma_b}{(D/2)}, \quad M = \frac{\sigma_b \times I}{D/2} = \frac{232.71 \times 2485}{7.5}$$

$$M = 77106 \text{ N/mm}.$$

$$\text{Moment} = P \times \delta = 1287.61 \times \delta = 77106$$

$$\boxed{\delta = 59.9 \text{ mm}}$$